(FP2) 4. Integration 专题作业1

NO. 1

OR

Let $I_n = \int_1^{\sqrt{2}} (x^2 - 1)^n \, dx$.

i. Show that, for $n \geq 1$, [5]

$$(2n + 1)I_n = \sqrt{2} - 2nI_{n-1}.$$

.....

ii. Using the substitution $x = \sec\theta$, show that [4]

$$I_n = \int_0^{\frac{1}{4}\pi} \tan^{2n+1}\theta \sec\theta \, d\theta.$$

.....

iii. Deduce the exact value of [5]

$$\int_0^{\frac{1}{4}\pi} \frac{\sin^7\theta}{\cos^8\theta} \, d\theta.$$

.....

A level FURTHER MATHEMATICS 2018/Winter/Paper1_2/Q11O

Mark scheme

Question	Answer	Marks	Guidance
i	$I_n = \left[x\left(x^2-1\right)^n\right]_1^{\sqrt{2}} - 2n\int_1^{\sqrt{2}} x^2\left(x^2-1\right)^{n-1}\,dx$	**M1A1**	Integrates by parts.
	$= \sqrt{2} - 2n\int_1^{\sqrt{2}} \left(x^2-1+1\right)\left(x^2-1\right)^{n-1}\,dx$	**M1**	Uses $x^2 = x^2 - 1 + 1$.
	$= \sqrt{2} - 2nI_n - 2nI_{n-1}$	**A1**	
	$\Rightarrow (2n+1)I_n = \sqrt{2} - 2nI_{n-1}$	**A1**	AG.
		5	
ii	$\frac{dx}{d\theta} = \tan\theta\sec\theta$	**M1A1**	Differentiates $\sec\theta$.
	$\sec\theta = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$ $\sec\theta = 1 \Rightarrow \theta = 0$	**B1**	Changes limits.
	$I_n = \int_0^{\frac{\pi}{4}} \left(\sec^2\theta - 1\right)^n \tan\theta\sec\theta\,d\theta = \int_0^{\frac{\pi}{4}} \tan^{2n+1}\theta\sec\theta\,d\theta$	**B1**	Uses $\sec^2\theta - 1 = \tan^2\theta$. (AG.)
		4	
iii	$\int_0^{\frac{\pi}{4}} \frac{\sin^7\theta}{\cos^8\theta}\,d\theta = I_3$	**B1**	Deduces that integral is I_3.
	$I_0 = \sqrt{2} - 1$ (or $I_1 = \frac{2-\sqrt{2}}{3}$)	**B1**	Calculates I_0 or I_1
	$3I_1 = \sqrt{2} - 2I_0 \Rightarrow I_1 = \frac{2-\sqrt{2}}{3}.$	**M1A1**	Uses reduction formula to find I_1 *or* I_2 ……
	$I_2 = \frac{\sqrt{2}}{5} - \frac{4}{5}I_1 = \frac{7\sqrt{2}-8}{15}$		Finds I_2.
	$I_3 = \frac{\sqrt{2}}{7} - \frac{6}{7}I_2 = \frac{16-9\sqrt{2}}{35}$	**A1**	Finds I_3.
		5	

Explanation

O.i

starting from I_n:

$$I_n = \int_1^{\sqrt{2}} (x^2-1)^n \, dx$$

$$= [x(x^2-1)^n]_1^{\sqrt{2}} - \int_1^{\sqrt{2}} x \cdot n(x^2-1)^{n-1} \cdot 2x \, dx$$ **M1A1**

(Integrate by parts)

$$= \sqrt{2} - 2n \int_1^{\sqrt{2}} x^2 (x^2-1)^{n-1} \, dx$$

(Try to construct I_{n-1} or I_n)

$$= \sqrt{2} - 2n \int_1^{\sqrt{2}} (x^2-1+1)(x^2-1)^{n-1} \, dx$$ **M1**

$$= \sqrt{2} - 2n \left(\int_1^{\sqrt{2}} (x^2-1)^n \, dx + \int_1^{\sqrt{2}} (x^2-1)^{n-1} \, dx \right)$$

$$= \sqrt{2} - 2n(I_n + I_{n-l})$$ **A1**

$$(2n+1)I_n = \sqrt{2} - 2nI_{n-1}$$ **A1**

ii)

x = secθ, $0 \leq \theta \leq \frac{\pi}{2}$

$$dx = \tan\theta \sec\theta \, d\theta$$ **M1A1**

when x=1, θ=0. When $x = \sqrt{2}, \ \theta = \frac{\pi}{4}$ **B1**

$$I_n = \int_0^{\frac{\pi}{4}} (\sec^2\theta - 1)^n \cdot \tan\theta \sec\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} (\tan^2\theta)^n \cdot \tan\theta \sec\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} tan^{2n+1}\theta \, sec\,\theta \, d\theta$$ **B1**

iii)

$$\int_0^{\frac{\pi}{4}} \frac{\sin^7\theta}{\cos^8\theta} d\theta = \int_0^{\frac{\pi}{4}} tan^7\theta \cdot sec\,\theta \, d\theta = I_3$$ **B1**

$$(2 \times 3 + 1)I_3 = \sqrt{2} - 2 \times 3 \cdot I_2$$

$$7I_3 = \sqrt{2} - 6I_2$$

$$5I_2 = \sqrt{2} - 4I_1$$

$$3I_1 = \sqrt{2} - 2I_0$$

$$I_0 = \int_1^{\sqrt{2}} 1 \, dx = \sqrt{2} - 1$$ **B1**

$I_1 = \frac{1}{3}\left(2-\sqrt{2}\right)$ **M1A1**

(Or $I_1 = \int_1^{\sqrt{2}} (x^2-1)\,dx = \left[\frac{1}{3}x^3 - x\right]_1^{\sqrt{2}} = \frac{2\sqrt{2}}{3} - \sqrt{2} - \frac{1}{3} + 1 = \frac{2}{3} - \frac{\sqrt{2}}{3}$)

$I_2 = \frac{1}{5}\left(\sqrt{2} - \frac{4}{3}\left(2-\sqrt{2}\right)\right) = \frac{1}{5}\left(\frac{7}{3}\sqrt{2} - \frac{8}{3}\right)$

$I_3 = \frac{1}{7}\left(\sqrt{2} - \frac{6}{5}\left(\frac{7}{3}\sqrt{2} - \frac{8}{3}\right)\right) = \frac{1}{7}\left(\frac{16}{5} - \frac{9}{5}\sqrt{2}\right) = \frac{16-9\sqrt{2}}{35}$ **A1**

Comment

OR:

i. Most candidates found the reduction formula, spotting the necessary split in the integrand.

ii. The second part was also well done, although a few candidates did not show the complete working, including finding the new limits.

iii. Candidates recognised that they needed to find I_3, and started by calculating either I_0 or I_1 correctly.

Answers: **(iii)** $I_3 = \frac{16-9\sqrt{2}}{35}$

NO. 2

Let $I_n = \int_0^1 (1+3x)^n e^{-3x}\,dx$, where n is an integer.

a. Show that $3I_n = 1 - 4^n e^{-3} + 3nI_{n-1}$. [3]

b. Find the exact value of I_2. [3]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_3/Q2

Mark scheme

Question	Answer	Marks
a	$I_n = \left[-\frac{1}{3}e^{-3x}(1+3x)^n\right]_0^1 + n\int_0^1 e^{-3x}(1+3x)^{n-1}\,dx$	M1 A1
	$-\frac{1}{3}e^{-3}4^n + \frac{1}{3} + nI_{n-1} \Rightarrow 3I_n = 1 - 4^n e^{-3} + 3nI_{n-1}$	A1
		3
b	$I_0 = \int_0^1 e^{-3x}\,dx = -\frac{1}{3}\left[e^{-3x}\right]_0^1 = \frac{1}{3}(1-e^{-3})$.	B1
	$I_1 = \frac{1}{3}(1 - 4e^{-3} + 1 - e^{-3}) = \frac{1}{3}(2 - 5e^{-3})$ $I_2 = \frac{1}{3}(1 - 16e^{-3} + 2(2 - 5e^{-3}))$	M1
	$I_2 = \frac{1}{3}(5 - 26e^{-3})$	A1
		3

Explanation

a.

Starting from I_n and integrate by parts:

$$I_n = \int_0^1 (1+3x)^n e^{-3x}\,dx$$

$$= \left[-\frac{1}{3}e^{-3x}(1+3x)^n\right]_0^1 - \int_0^1 \left(-\frac{1}{3}e^{-3x}\right) n(1+3x)^{n-1}\cdot 3\,dx$$ **M1 A1**

$$= \left(-\frac{1}{3}e^{-3}(1+3)^n + \frac{1}{3}(1+0)^n\right) + \int_0^1 ne^{-3x}(1+3x)^{n-1}\,dx$$

$$= \frac{1}{3} - \frac{1}{3}e^{-3}4^n + nI_{n-1}$$

$$3I_n = 1 - e^{-3}4^n + 3nI_{n-1}$$ **A1**

b)

$$I_0 = \int_0^1 e^{-3x}\,dx = \left[-\frac{1}{3}e^{-3x}\right]_0^1 = -\frac{1}{3}e^{-3} + \frac{1}{3}$$ **B1**

When n=1

$3I_1 = 1 - e^{-3}4 + 3I_0 = 1 - 4e^{-3} + 1 - e^{-3} = 2 - 5e^{-3}$

When n=2

$3I_2 = 1 - e^{-3}4^2 + 6I_1 = 1 - 16e^{-4} + 2(2 - 5e^{-3}) = 5 - 26e^{-3}$ **M1**

$I_2 = \frac{1}{3}(5 - 26e^{-3})$ **A1**

NO. 3

The integral I_n, where n is an integer, is defined by $I_n = \int_0^{\frac{1}{2}} \left(1-x^2\right)^{-\frac{1}{2}n} \mathrm{d}x$.

a. Find the exact value of I_1. [2]

b. By considering $\frac{\mathrm{d}}{\mathrm{d}x}\left(x\left(1-x^2\right)^{-\frac{1}{2}n}\right)$, or otherwise, show that $nI_{n+2} = 2^{n-1}3^{-\frac{1}{2}n} + (n-1)I_n$. [5]

c. Find the exact value of I_5 giving the answer in the form $k\sqrt{3}$, where k is a rational number to be determined. [3]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_1/Q6

Mark scheme

Question	Answer	Marks
a	$I_1 = \int_0^{\frac{1}{2}} \left(1-x^2\right)^{-\frac{1}{2}} \mathrm{d}x = \left[\sin^{-1} x\right]_0^{\frac{1}{2}} = \frac{1}{6}\pi$	M1 A1
		2
b	$\frac{\mathrm{d}}{\mathrm{d}x}\left(x\left(1-x^2\right)^{-\frac{1}{2}n}\right) = nx^2\left(1-x^2\right)^{-\frac{1}{2}n-1} + \left(1-x^2\right)^{-\frac{1}{2}n}$	M1 A1
	$= n\left(1-\left(1-x^2\right)\right)\left(1-x^2\right)^{-\frac{1}{2}n-1} + \left(1-x^2\right)^{-\frac{1}{2}n}$	M1
	$\left[x\left(1-x^2\right)^{-\frac{1}{2}n}\right]_0^{\frac{1}{2}} = nI_{n+2} - nI_n + I_n$	M1
	$\frac{1}{2}\left(\frac{3}{4}\right)^{-\frac{1}{2}n} = nI_{n+2} - (n-1)I_n \Rightarrow nI_{n+2} = 2^{n-1}3^{-\frac{1}{2}n} + (n-1)I_n$	A1
		5
c	$I_3 = 3^{-\frac{1}{2}}$	B1
	$3I_5 = 2^2 3^{-\frac{3}{2}} + 2I_3 \Rightarrow I_5 = \frac{10}{27}\sqrt{3}$	M1 A1
		3

Explanation

a.

Use the result in formula sheet:

$$I_1 = \int_0^{\frac{1}{2}} (1-x^2)^{-\frac{1}{2}}\,dx = \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}dx = [\sin^{-1} x]_0^{\frac{1}{2}} = \frac{\pi}{6} - 0 = \frac{\pi}{6}$$ **M1 A1**

b)

$$\frac{d}{dx}\left(x(1-x^2)^{-\frac{1}{2}n}\right) = (1-x^2)^{-\frac{1}{2}n} + x\left(-\frac{1}{2}n\right)(1-x^2)^{-\frac{1}{2}n-1}(-2x)$$ **M1 A1**

$$= (1-x^2)^{-\frac{1}{2}n} + nx^2(1-x^2)^{-\frac{n+2}{2}}$$

Integrate:

$$x(1-x^2)^{-\frac{1}{2}n} = \int (1-x^2)^{-\frac{1}{2}n}dx + n\int x^2(1-x^2)^{-\frac{n+2}{2}}dx$$

$$I_n = \left[x(1-x^2)^{-\frac{1}{2}n}\right]_0^{\frac{1}{2}} - n\int_0^{\frac{1}{2}} x^2(1-x^2)^{-\frac{n+2}{2}}dx$$

$$= \frac{1}{2}\left(1-\frac{1}{2^2}\right)^{-\frac{1}{2}n} + n\int_0^{\frac{1}{2}}(1-x^2-1)(1-x^2)^{-\frac{n+2}{2}}dx$$ **M1**

$$= \frac{1}{2}\left(\frac{3}{4}\right)^{-\frac{1}{2}n} + n\int_0^{\frac{1}{2}}(1-x^2)(1-x^2)^{-\frac{n+2}{2}}dx - n\int_0^{\frac{1}{2}}(1-x^2)^{-\frac{n+2}{2}}dx$$

$$= \frac{1}{2}4^{\frac{1}{2}n}3^{-\frac{1}{2}n} + n\int_0^{\frac{1}{2}}(1-x^2)^{-\frac{n}{2}}dx - n\int_0^{\frac{1}{2}}(1-x^2)^{-\frac{n+2}{2}}dx$$

$$= 2^{n-1}3^{-\frac{1}{2}n} + nI_n - nI_{n+2}$$ **M1**

So $nI_{n+2} = 2^{n-1}3^{-\frac{1}{2}n} + nI_n - I_n = 2^{n-1}3^{-\frac{1}{2}n} + (n-1)I_n$ **A1**

c)

when n=1

$$I_3 = 2^0 \cdot 3^{-\frac{1}{2}} + 0 \cdot I_1 = \frac{1}{\sqrt{3}}$$ **B1**

When n=3

$$3I_5 = 2^2 3^{-\frac{3}{2}} + 2I_3 = \frac{4}{3\sqrt{3}} + \frac{2}{\sqrt{3}} = \frac{10}{3\sqrt{3}} = \frac{10\sqrt{3}}{9}$$

$$I_5 = \frac{10\sqrt{3}}{27}$$ **M1 A1**

NO. 4

It is given that, for $n \geq 0$,

$$I_n = \int_0^1 x^n e^{x^3}\, dx.$$

i. Show that $I_2 = \frac{1}{3}(e - 1)$. [2]

.....

ii. Show that, for $n \geq 3$, [3]

$$3I_n = e - (n-2)I_{n-3}.$$

.....

iii. Hence find the exact value of I_8. [3]

.....

A level FURTHER MATHEMATICS 2019/Summer/Paper1_1/Q4

Mark scheme

Question	Answer	Marks	Guidance
4.i	$\int_0^1 x^2 e^{x^3}\, dx = \frac{1}{3}\left[e^{x^3}\right]_0^1 = \frac{1}{3}(e-1)$	**M1 A1**	Must show working, AG.
		2	
ii	$I_n = \int_0^1 x^{n-2} x^2 e^{x^3}\, dx = \left[\frac{1}{3}x^{n-2}e^{x^3}\right]_0^1 - \frac{n-2}{3}\int_0^1 x^{n-3}e^{x^3}\, dx$	**M1 A1**	Integrates by parts.
	$\frac{e}{3} - \frac{n-2}{3}I_{n-3} \Rightarrow 3I_n = e - (n-2)I_{n-3}$	**A1**	AG
		3	
iii	$I_5 = \frac{1}{3}(e - 3I_2) = \frac{1}{3}(e - e + 1) = \frac{1}{3}$	**M1 A1**	Applies reduction formula once and uses **(i)**.
	$I_8 = \frac{1}{3}(e - 6I_5) = \frac{1}{3}(e-2).$	**A1**	Must be exact.
		3	

Explanation

4.i

Notice that $\left(e^{x^3}\right)' = 3x^2 e^{x^3}$

$I_2 = \int_0^1 x^2 e^{x^3}\,dx = \frac{1}{3}\left[e^{x^3}\right]_0^1 = \frac{1}{3}(e-1)$ **M1 A1**

ii)

try to use I_n to find a expression in terms of I_{n-3}:

$I_n = \int_0^1 x^n e^{x^3}\,dx = \left[\frac{1}{3}x^{n-2}e^{x^3}\right]_0^1 - \int_0^1 \frac{1}{3}e^{x^3}(n-2)x^{n-3}\,dx$ (Integrate by parts) **M1 A1**

$= \left(\frac{1}{3}\cdot 1\cdot e - 0\right) - \frac{1}{3}(n-2)\int_0^1 x^{n-3}e^{x^3}\,dx$

$= \frac{e}{3} - \frac{1}{3}(n-2)I_{n-3}$

$3I_n = e - (n-2)I_{n-3}$ **A1**

iii)

Use the equation in ii)

When n=5

$3I_5 = e - 3I_2 = e - 3\left(\frac{1}{3}(e-1)\right) = 1$

$I_5 = \frac{1}{3}$ **M1 A1**

When n=8$3I_8 = e - 6I_5$=$e - 6\times\frac{1}{3} = e-2$

$I_8 = \frac{1}{3}(e-2)$ **A1**

Comment

i. Many candidates correctly used the idea of reverse differentiation. Reponses which involved the approach of integration by parts were less successful.
ii. Almost all candidates applied integration by parts to I_n. Those who used the result from the previous part, or integrated x^n, successfully derived the given reduction formula. Others who tried to integrate e^{x^3} were less successful.
iii. This part was well done with the majority of candidates accurately applying the reduction formula.

NO. 5

Let $I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \, dx$ for $n > 0$.

i. Find the value of I_2. [2]

.....

ii. Show that, for $n > 2$, [5]

$$(n-1)I_n = 2^{\frac{1}{2}n-1} + (n-2)I_{n-2}.$$

.....

iii. The curve C has equation $y = \sec^3 x$ for $0 \leq x \leq \frac{1}{4}\pi$. The region R is bounded by C, the x-axis, the y-axis and the line $x = \frac{1}{4}\pi$. Find the volume of revolution generated when R is rotated through 2π radians about the x-axis. [4]

.....

A level FURTHER MATHEMATICS 2017/Winter/Paper1_1/Q8

Mark scheme

Question	Answer	Marks	Gui
8.i	$I_2 = \int_0^{\frac{1}{4}\pi} \sec^2 x \mathrm{d}x = \left[\tan x\right]_0^{\frac{1}{4}\pi} = 1$	M1A1	
		2	
ii	$I_n = \int_0^{\frac{1}{4}\pi} \sec^{n-2} x.\sec^2 x \mathrm{d}x$	M1	
	$= \left[\sec^{n-2} x \tan x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} (n-2)\sec^{n-3} x(\sec x \tan x)\tan x \mathrm{d}x$	M1A1	
	$= \left[\sec^{n-2} x \tan x\right]_0^{\frac{1}{4}\pi} - (n-2)\int_0^{\frac{1}{4}\pi} \sec^{n-2} x\left(\sec^2 x - 1\right)\mathrm{d}x$	M1A1	
	$\Rightarrow (n-1)I_n = 2^{\frac{1}{2}n-1} + (n-2)I_{n-2}$		AG
		5	
iii	Volume of revolution $= \pi\int y^2 \mathrm{d}x = \pi\int_0^{\frac{1}{4}\pi} \sec^6 x \mathrm{d}x$	M1	
	$3I_4 = 2 + 2\times 1 \Rightarrow I_4 = \frac{4}{3}$	M1	
	$5I_6 = 4 + 4\times\frac{4}{3} \Rightarrow I_6 = \frac{28}{15}$	M1	
	Volume of revolution $= \frac{28\pi}{15}$	A1	
		4	

Explanation

8.i

$I_2 = \int_0^{\frac{1}{4}\pi} \sec^2 x \mathrm{d}x = \left[\tan x\right]_0^{\frac{1}{4}\pi} = 1$ **M1A1**

8.ii

$$I_n = \int_0^{\frac{1}{4}\pi} \sec^{n-2}x.\sec^2 x dx$$ **M1**

$$= [\sec^{n-2} x \tan x]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan x\,(n-2)\sec^{n-3} x\,(\sec x \tan x)\,dx$$ **M1A1**

$$= \left(\sqrt{2}\right)^{n-2} - (n-2)\int_0^{\frac{\pi}{4}} tan^2 x\, sec^{n-2} x\, dx$$

$$= 2^{\frac{n}{2}-1} - (n-2)\int_0^{\frac{\pi}{4}} (sec^2 x - 1)\, sec^{n-2} x\, dx$$ **M1A1**

$$= 2^{\frac{n}{2}-1} - (n-2)(\int_0^{\frac{\pi}{4}} sec^2 x\, sec^{n-2} x\, dx + \int_0^{\frac{\pi}{4}} sec^{n-2} x\, dx)$$

$$= 2^{\frac{n}{2}-1} - (n-2)(I_n - I_{n-2})$$

$$(n-1)I_n = 2^{\frac{1}{2}n-1} + (n-2)I_{n-2}$$

9.iii

volume of revolution is given by: $V = \int \pi y^2\, dx$

$$V = \int_0^{\frac{\pi}{4}} \pi\, sec^6 x\, dx = \pi \cdot I_6$$ **M1**

Use the result in ii)

n=6 $(6-1)I_6 = 2^{3-1} + (6-2)I_4 \rightarrow I_6 = \frac{1}{5}(4 + 4I_4)$

n=4

$$(4-1)I_4 = 2^{2-1} + (4-2)I_2 \rightarrow I_4 = \frac{1}{3}(2 + 2I_2) = \frac{1}{3}(2+2) = \frac{4}{3}$$ **M1**

$$I_6 = \frac{1}{5}(4 + 4I_4) = \frac{1}{5}\left(4 + \frac{16}{3}\right) = \frac{28}{15}$$ **M1**

$$V = \pi \cdot I_6 = \frac{28\pi}{15}$$ **A1**

Comment

i. Most candidates remembered how to integrate this function and almost all showed the function as necessary before substituting limits.

ii. The majority of candidates split the integrand and used integration by parts correctly. Some did not justify the value $2^{\frac{1}{2}n-1}$ but most made each step very clear on the way to the given answer. It proved easier to work with secθ and tanθ than revert to cosθ and sinθ, but other methods were used successfully too.

iii. Those who remembered the formula correctly did very well on this part, others were still able to pick up marks for using the given reduction formula. Others embarked on complicated integrations that did not use the result of **part (ii)** as they should have expected.

Answers: **(i)** 1 **(iii)** $\frac{28\pi}{15}$

NO. 6

i. Let $I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \sin^2 x \, dx$, for $n \geq 0$. By differentiating $\cos^{n-1}x\sin^3 x$ with respect to x, prove that [5]

$$(n+2)I_n = (n-1)I_{n-2} \quad \text{for } n \geqslant 2.$$

ii. Hence find the exact value of I_4. [4]

A level FURTHER MATHEMATICS 2016/Summer/Paper1_1/Q5

Mark scheme

Qu	Solution	Part Marks
5.i	$\frac{d}{dx}\left(\cos^{n-1}x\sin^3 x\right) = -(n-1)\cos^{n-2}x\sin^4 x + 3\cos^n x\sin^2 x$	M1A1
	$\Rightarrow \left[\cos^{n-1}x\sin^3 x\right]_0^{\frac{1}{2}\pi} = -\int_0^{\frac{1}{2}\pi}(n-1)\cos^{n-2}x\sin^2 x\left(1-\cos^2 x\right)dx + 3I_n$	M1
	$\Rightarrow$ 0 = $-(n-1)I_{n-2} + (n-1)I_n + 3I_n$	M1
	$\Rightarrow$ = $(n+2)I_n = (n-1)I_{n-2}$ **(AG)**	A1
		[5]
ii	$I_0 = \int_0^{\frac{1}{2}\pi}\sin^2 x dx = \frac{1}{2}\int_0^{\frac{1}{2}\pi}(1-\cos 2x)dx$	M1
	$= \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\frac{1}{2}\pi} = \frac{\pi}{4}$	A1
	$I_2 = \frac{1}{4}\times\frac{\pi}{4}I_4 = \frac{1}{2}\times\frac{1}{4}\times\frac{\pi}{4} = \frac{\pi}{32}$	M1A1
		[4]

Explanation

5.i

$\frac{d}{dx}(\cos^{n-1}x\sin^3 x)$

$= (n-1)\cos^{n-2}x(-\sin x)\sin^3 x + 3\sin^2 x\cos x\cos^{n-1}x$

$= -(n-1)\sin^4 x\cos^{n-2}x + 3\sin^2 x\cos^n x$ **M1A1**

Use the result above

$$3I_n = [cos^{n-1} x \, sin^3 x]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \, sin^4 x \, cos^{n-2} x \, dx$$

$$= (0-0) + (n-1)\int_0^{\frac{\pi}{2}} (1-\cos^2 x)\sin^2 x \cos^{n-2} x \, dx$$

$$= (n-1)\int_0^{\frac{\pi}{2}} \sin^2 x \cos^{n-2} x \, dx - (n-1)\int_0^{\frac{\pi}{2}} \sin^2 x \cos^n x \, dx$$ **M1**

$$= (n-1)I_{n-2} - (n-1)I_n$$ **M1**

$$(n+2)I_n = (n-1)I_{n-2}$$ **A1**

ii.

$$I_0 = \int_0^{\frac{\pi}{2}} sin^2 x \, dx = \int_0^{\frac{\pi}{2}} \frac{1-cos 2x}{2} dx = \left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}}$$

use double angle formula. M1

$$= \frac{\pi}{4} - 0 - 0 - 0 = \frac{\pi}{4}$$ **A1**

When n=2

$$4I_2 = I_0 \rightarrow I_2 = \frac{1}{4} I_0 = \frac{\pi}{16}$$

When n=4

$$6I_4 = 3I_2 \rightarrow I_4 = \frac{1}{2} I_2 = \frac{\pi}{32}$$

M1A1

Comment

Where candidates used the given differentiation as instructed, they were usually able to spot the necessary manipulation of $\cos^2 x$ and complete the proof. However, some candidates left out steps of working and, because the answer was given, they lost marks. The majority of students successfully found I_0, and others used trigonometric manipulation to calculate I_2.

Answers: **(ii)** $I_4 = \frac{\pi}{32}$

NO. 7

Find the exact value of $\int_2^{\frac{7}{2}} \frac{1}{\sqrt{4x-x^2-1}}\,\mathrm{d}x$. [5]

A level FURTHER MATHEMATICS 2024/Summer/Paper2_3/Q1

Mark scheme

Question	Answer	Marks	Guidance
1	$-x^2 + 4x - 1 = 3 - (x-2)^2$	B1	Completes the square.
	$\int_2^{\frac{7}{2}} \frac{1}{\sqrt{3-(x-2)^2}}\,\mathrm{d}x = \left[\sin^{-1}\left(\frac{x-2}{\sqrt{3}}\right)\right]_2^{\frac{7}{2}}$	M1 A1	Applies formula. $\left[\sin^{-1}\left(\frac{x-2}{\sqrt{3}}\right)\right]_2^{\frac{7}{2}} = \left[-\cos^{-1}\left(\frac{x-2}{\sqrt{3}}\right.\right.$
	$\sin^{-1}\left(\frac{1}{2}\sqrt{3}\right) - \sin^{-1}(0) = \frac{1}{3}\pi$	M1 A1	Uses limits. Answer must be in radians.
		5	

Comment

Most candidates realised that they needed to complete the square and were usually successful. A few other less efficient methods were seen involving elaborate changes of variable. A small minority of candidates lost the final mark due to not evaluating the inverse sine expression.

NO. 8

a. Starting from the definitions of tanh and sech in terms of exponentials, prove that

$$1 - \tanh^2 x = \operatorname{sech}^2 x.$$ [3]

b. Using the substitution $u = \tanh x$, or otherwise, find $\int \operatorname{sech}^2 x \tanh^2 x \, dx$. [2]

It is given that, for $n \geqslant 0$, $I_n = \int_0^{\ln 3} \operatorname{sech}^n x \tanh^2 x \, \mathrm{d}x$.

c. Show that, for $n \geqslant 2$,

$$(n+1)I_n = \left(\frac{4}{5}\right)^3 \left(\frac{3}{5}\right)^{n-2} + (n-2)I_{n-2}.$$ [5]

[You may use the result that $\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.]

d. Find the value of I_4. [3]

A level FURTHER MATHEMATICS 2021/Winter/Paper2_2/Q8

Mark scheme

Question	Answer	Marks	Guidance
a	$\tanh x = \dfrac{e^x - e^{-x}}{e^x + e^{-x}} \quad \operatorname{sech} x = \dfrac{2}{e^x + e^{-x}}$	B1	Writes in exponential form.
	$1 - \left(\dfrac{e^x - e^{-x}}{e^x + e^{-x}}\right)^2 = \dfrac{\left(e^x + e^{-x}\right)^2 - \left(e^x - e^{-x}\right)^2}{\left(e^x + e^{-x}\right)^2} = \dfrac{4}{\left(e^x + e^{-x}\right)^2}$	M1	Writes over common denominator.
	$\operatorname{sech}^2 x$	A1	AG
		3	
b	$\int u^2 \, du = \frac{1}{3}\tanh^3 x + C$	M1 A1	Uses substitution correctly. Allow by inspection.
		2	
c	$I_n = \int_0^{\ln 3} \operatorname{sech}^{n-2} x \operatorname{sech}^2 x \tanh^2 x \, dx$	B1	Separates into correct structure.
	$\left[\frac{1}{3}\operatorname{sech}^{n-2} x \tanh^3 x\right]_0^{\ln 3} + \frac{1}{3}(n-2)\int_0^{\ln 3} \operatorname{sech}^{n-2} x \tanh^4 x \, dx$	M1	Uses integration by parts correctly.
	$\frac{1}{3}\left(\frac{3}{5}\right)^{n-2}\left(\frac{4}{5}\right)^3 + \frac{1}{3}(n-2)\left(I_{n-2} - I_n\right)$	M1 A1	Uses $1 - \operatorname{sech}^2 x = \tanh^2 x$.
	$(3+n-2)I_n = \left(\frac{3}{5}\right)^{n-2}\left(\frac{4}{5}\right)^3 + (n-2)I_{n-2}$ leading to $(n+1)I_n = \left(\frac{3}{5}\right)^{n-2}\left(\frac{4}{5}\right)^3 + (n-2)I_{n-2}$	A1	AG
		5	
d	$I_2 = \frac{1}{3}\left(\frac{4}{5}\right)^3 = \frac{64}{375}$	B1	
	$(4+1)I_4 = \left(\frac{3}{5}\right)^2\left(\frac{4}{5}\right)^3 + 2I_2$ leading to $I_4 = \dfrac{4928}{46875} = 0.105$	M1 A1	Applies reduction formula with $n = 4$.
		3	

Comment

a. This part of the question was well done with the majority of candidates accurately substituting tanh and sech in terms of exponentials and writing over a common denominator to justify the given identity.
b. The majority of candidates used the given substitution correctly. A few did not add an arbitrary constant to their answer.
c. Most candidates separated the integral into the correct parts, using **part (b)**. Some candidates maintained accuracy throughout, clearly applying $\tanh^2 x = 1 - \operatorname{sech}^2 x$ to derive the given reduction formula.
d. This part was well done with most candidates accurately applying the reduction formula.